SOLID MECHANICS FOR AERONAUTICS

III Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE03	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

UNIT-I	SIMPLE STRESSES AND STRAINS

SIMPLE STRESSES AND STRAINS: Elasticity and plasticity, Types of stresses and strains, Hooke's law, stress-strain diagram for mild steel, Working stress, Factor of safety, Lateral strain, Poisson's ratio and volumetric strain, Elastic constants and the relationship between them, stepped bars, composite bars, Temperature stresses. Strain energy-suddenly applied and impact loads

UNIT-II	SHEAR FORCE AND BENDING MOMENT

SHEAR FORCE AND BENDING MOMENT: Introduction to beam, Shear force and bending moment, Relation between Shear Force and Bending Moment. Shear Force and Bending Moment diagrams for cantilever, simply supported and overhanging beams subjected to point loads and U.D.L., Point of contra flexure.

UNIT-III	REDUNDANT STRUCTURES

DEFLECTION OF BEAMS: Double integration methods, Determination of slope and deflection for cantilever and simply supported beams subjected to point loads. Indeterminate structure**,** order of redundancy, moment area method, analysis of various types of beams, three moment equation (Clapeyron theorem) of a continuous beam.Principle of virtual Displacements and principle of Virtual Force Castigliano's theorems, maxwell's reciprocal theorem and unit load method

UNIT-IV	FLEXURAL STRESSES and TORSIONAL STRESSES

FLEXURAL STRESSES: Theory of simple bending, Assumptions, Derivation of bending equation: $M/I = f/y = E/R$, Neutral axis, Determination bending stresses-I, T sections- section modulus of rectangular and circular sections (Solid and Hollow),
TORSION OF CIRCULAR SHAFTS: Theory of pure torsion - derivation of Torsion equations: $T/J = q/r = G\theta/L$ - Assumptions made in the theory of pure torsion - Torsional moment of resistance - Polar section modulus - Power transmitted by shafts

UNIT-V	PRINCIPAL STRESSES AND STRAINS & FAILURE THEORIES

PRINCIPAL STRESSES AND STRAINS: Introduction - Stresses on an inclined section of a bar under axial loading - compound stresses - Normal and tangential stresses on an inclined plane for biaxial stresses - Two perpendicular normal stresses accompanied by a state of simple shear - Mohr's circle of stresses
Various theories of failure - Maximum Principal Stress Theory, Maximum Principal Strain Theory, Strain Energy and Shear Strain Energy Theory (Von Mises Theory).

Text Books:

1.Ramamrutham. S (2012), Strength of materials, 17th edition, Dhanpat Rai Publications, Engineering Mechanics/Timoshenko and D.H. Young, Mc Graw Hill Book Company New Delhi, India.
2.Dr.Bansal R.K*(2007),* Strength of materials, 10th edition,Laxmi Publications,Hyderabad

Reference Books:

1. Ryder G. H (2007), Strength of materials, 3rd edition, Macmillan, New Delhi, India.
2. Bhavikathi S. S (2008), Strength of materials, 3rd edition, Vikas Publishing House, New Delhi, India

COURSE OUTCOMES: STUDENTS SHOULD BE ABLE TO:

1. Apply the concept of stress and strain to analyze and design structural members
2. Develop the shear force and bending moment diagrams for different beams subjected to various loads.
3. Determine deflection and slop for determinant and redundant structures.
4. Design the circular shafts and Compute the bending stresses.
5. Compute the principal stresses and analyse the elastic deformation of members by apply various failure theories.